Project Report

Ring Oscillator based Truly Random Number Generator in Cadence Virtuoso

Submitted By

S. No.	Team Member	Roll No. / Batch
1	Berson Meshak J	23294917001 (ECE A – B1)
2	Nikhil Yadav	23294917151 (ECE B – B2)
3	Sarvesh Kumar	23294917080 (ECE B – B1)
4	Kushi Dixit	23294917035 (ECE B – B1)

B. Tech in Electronics and Communication Engineering
(Semester V)

Under the Supervision and Co-Supervision

Prof. P. K. Singh **Dr. Khushwant Sehra**

FACULTY OF TECHNOLOGY
UNIVERSITY OF DELHI
NEW DELHI – 110007
(2025 – 2026)

Certificate

This is to certify that the project entitled **"Ring Oscillator based Truly Random Number Generator design in Cadence Virtuoso"** was successfully completed by the following students.

1.	Berson Meshak J	Faculty of Technology, University of Delhi
2.	Nikhil Yadav	Faculty of Technology, University of Delhi
3.	Sarvesh Kumar	Faculty of Technology, University of Delhi
4.	Kushi Dixit	Faculty of Technology, University of Delhi

We hereby acknowledge the mentorship and guidance provided by **Prof. P.K. Singh**, **Dr. Khushwant Sehra**, whose continuous support and expert insights were instrumental in the successful completion of this project.

Prof. P.K.Singh

Supervisor

Department of Electronics and Communication Engineering

Faculty of Technology, University of Delhi

(Signature)

Dr.Khushwant Sehra

Co-Supervisor

Department of Electronics and Communication Engineering

Faculty of Technology, University of Delhi

(Signature)

Declaration

We hereby declare that the project work presented in this report, entitled **"Ring Oscillator based Truly Random Number Generator design in Cadence Virtuoso"** is entirely our own work and has not been submitted for any degree or diploma from this or any other institute for the partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology (B.Tech)**.

S. No.	**Team Member**	**Department / Institute**	**Signature**
1	Berson Meshak J	Faculty of Technology, University of Delhi	
2	Nikhil Yadav	Faculty of Technology, University of Delhi	
3	Sarvesh Kumar	Faculty of Technology, University of Delhi	
4	Kushi Dixit	Faculty of Technology, University of Delhi	

Acknowledgement

We would like to express our sincere gratitude to our college, Faculty of Technology, University of Delhi, for providing us with the opportunity to undertake this project. Their continuous encouragement and support have played a crucial role in our academic and personal growth.

We are especially thankful to the members of our project team: **Berson Meshak J, Nikhil Yadav, Sarvesh Kumar, Kushi Dixit** for their unwavering support, dedication, and collaborative spirit. It has been a rewarding and enriching experience working alongside each one of you.

Our deepest appreciation goes to our Supervisor and Co-Supervisor **Prof.P.K.Singh, Dr.Khushwant Sehra**, for their invaluable guidance, continuous support, and constructive feedback throughout the duration of this project. Their mentorship has been instrumental in shaping our work and driving it to completion.

Lastly, we are sincerely grateful to all the faculty members of our department who have contributed significantly to our overall education and development.

Abstract

This project presents the design and simulation of a True Random Number Generator (TRNG) utilising the intrinsic stochastic properties of CMOS transistors. The main entropy source is a high-speed **Ring Oscillator** made from an odd number of inverter stages, implemented at the transistor level in the **Cadence Virtuoso** envnment using the **gpdk90** technology lib.

The fundamental randomness is derived from timing jitter inherent in the RO's oscillation, a process arising from unavoidable noise sources like thermal noise and flicker noise. This **jitter-laden signal** is sampled by a carefully designed **D-Flip Flop (DFF)** clocked by a stable and external reference frequency.

The sampling process deliberately exploits metastability, converting analog uncertainty into a digital bits of 1s and 0s.Extensive transient simulations using the Spectre simulator confirms the circuit's functionality, demonstrating an aperiodic-unpredictable output sequence.

The report details the full-fledged design flow from schematic capture and simulation to analysis, successfully evaluating and provig a hardware-efficient TRNG architecture suitable for cryptographic systems, secure key generation, and other applications requiring unpredictable randomness.

With successful schematic verification, the complete-custom layout of the TRNG was carefully created, incorporating techniques to alleviate proximity and layout-dependent effects. The design has undergone **Design Rule Checking (DRC)** and **Layout Versus Schematic (LVS)** verification to ensure functional correctness. **layout simulations** were performed majorly to quantify the impact of parasitic extraction on both performance and randomness.

The report expalins the complete IC design flow from schematic capture and simulation to physical and post-layout validation, successfully demonstrating a hardware-efficient TRNG architecture suitable for cryptographic systems, secure key generation, and other applications requiring unpredicatble randomness.

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Introduction

Security and trust have become integral components of new electronic design as a result of growing dependence on digital systems. Whereas cryptographic mechanisms protect our data, allow for secure communication, and ensure the integrity of connected devices, one of the very foundational needs for the processes is high-quality random numbers that form the basics for encryption keys and authentication processes.

Weak, predictable randomness directly compromises the system security, making the design of reliable entropy sources a major critical challenge.

Random number generators are typic classified into Pseudo-Random Number Generators (PRNGs) and True Random Number Generators (TRNGs). PRNGs are algorithmic and optimised but remain predictable if their internal state or maybe seed is exposed. In contrast, TRNGs extract randomness from physical and randombread processes, providing inherently unpredictable output suitable for secure hardware applications.

Among hardware-based TRNG architectures, ring-oscillator designs are most commonly because of their simple, less area, and compatibility with standard CMOS technology. These oscillators possess natural variations such as phase noise and jitter, which can be used as entropy sources.

This project focuses on the design and working of a Ring Oscillator. The work describes the full process starting—from transistor-level design and simulation to sampling circuit integration and full layout development— resulting in a DRC- and LVS-clean design ready for its fabrication. The aim is to demonstrate a strong hardware entropy source suitable for secure integrated circuit applications. we take the circuit from the Reference paper **"S. Williams, P. Kirtonia, S. Akter, K. Khalil, and M. Bayoumi, “A Novel Pentagonal Ring Oscillator as a True Random Number Generator,” in 2025 IEEE 18th Dallas Circuits and Systems Conference (DCAS), 2025, pp. 1–5"**.

- This is the design and transistor-level simulation of a high-gain ring oscillator.
- This is the integration of a sampling element (D-Flip-Flop) and a stable ref. clock.
- Ths is a complete analysis of the output bitstream for randomness.
- This is the creation of a full-custom, DRC and LVS clean layout, ready for fabrication.

1. Architectural Overview

The design and enhancement of the Ring Oscillator (RO) for the True Random Number Generator (TRNG) followed a structured, step-by-step method. The primary goal was to progressively develop and optimise the oscillator circuit to achieve efficient performance metrics, specially targeting a high oscillation frequency and favorable spectral characteristics (measured through Total Harmonic Distortion - THD) to maximize timing jitter for entropy retrieval. The architecture and design flow is as follows.

1.1 Design and Analysis of the Standard CMOS Inverter

- We begin the project with the design and characterization of a basic, symetrical CMOS inverter in Cadence.
- This primarily cell was optimized for a practical rise and fall time and served as the theoretical building block for the starting oscillator stages.

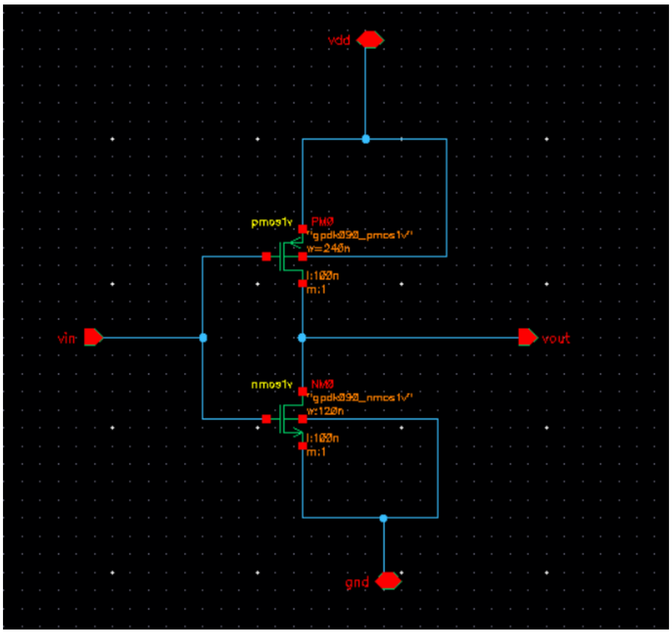

Figure 1.1: CMOS Inverter Schematic

Analysis: The primary inverter gives a robust switching behavior fueled by transistor sizing (W/L). The rise and fall symmetry was tuned to achieve homogenous propagation delay.

- The CMOS inverter was made with correct match and diffusion.

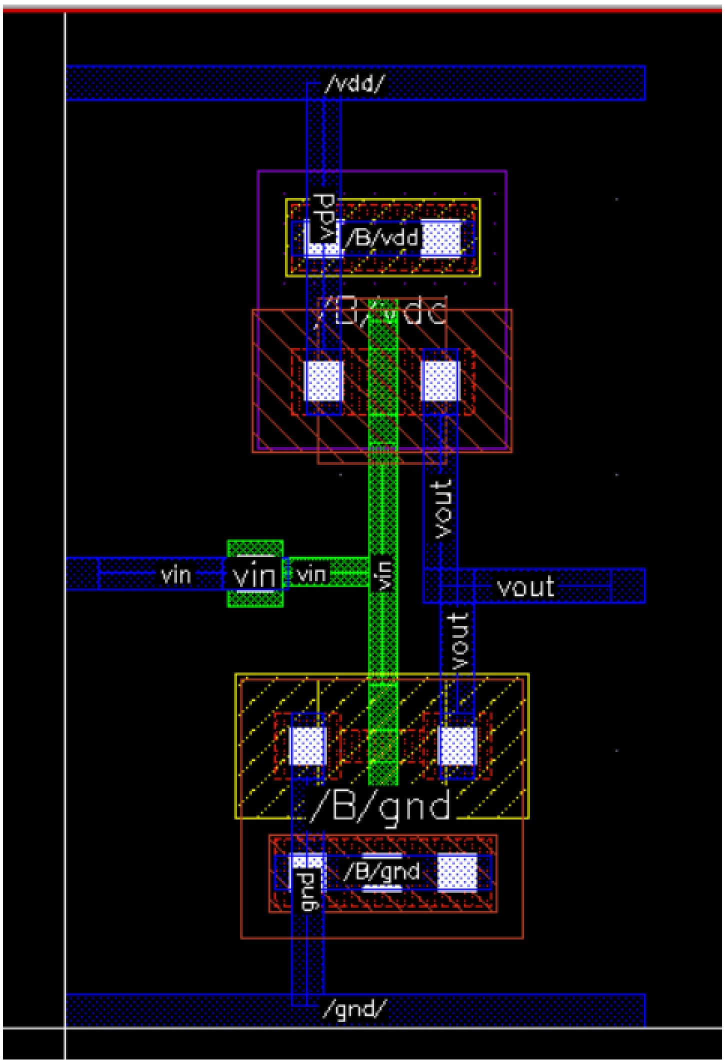

Figure 1.2: Layout of the CMOS Inverter

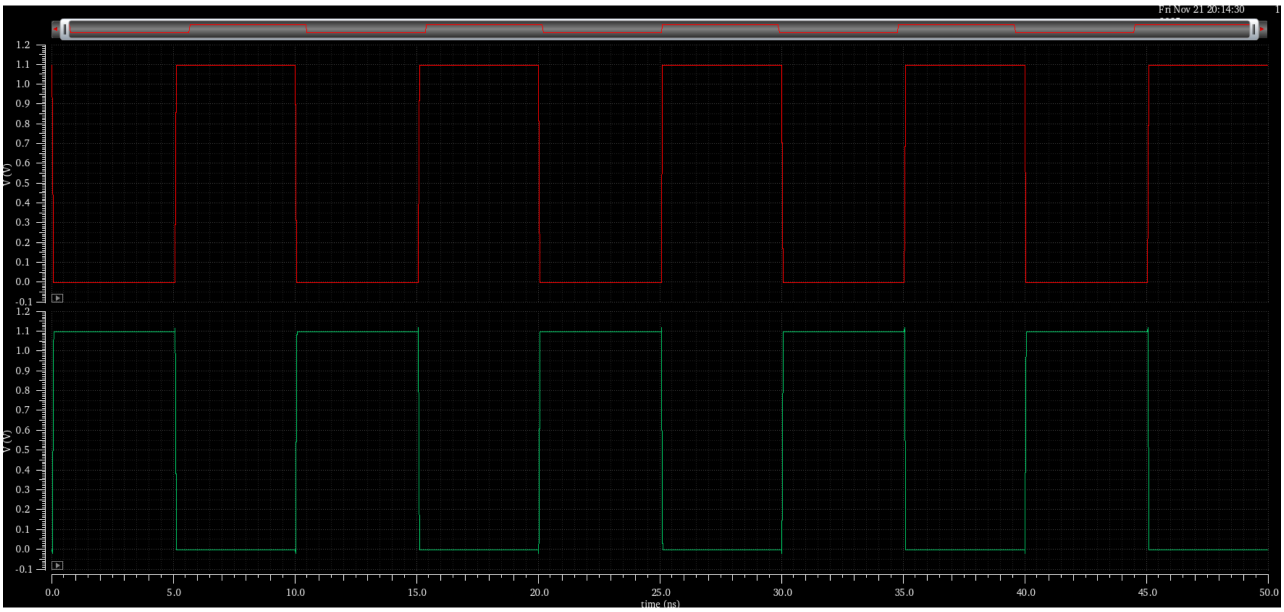

Figure 1.3: Transient Response of CMOS Inverter

1.2 Design and Implementation of the CMOS Ring Oscillator

The RO WORKS by connecting an odd number of CMOS inverters in a closed loop.Signal everytime toggles between logic levels due to propogation delay; producing sustained oscillations.We have high oscillation freq. when we connect least number of stages.

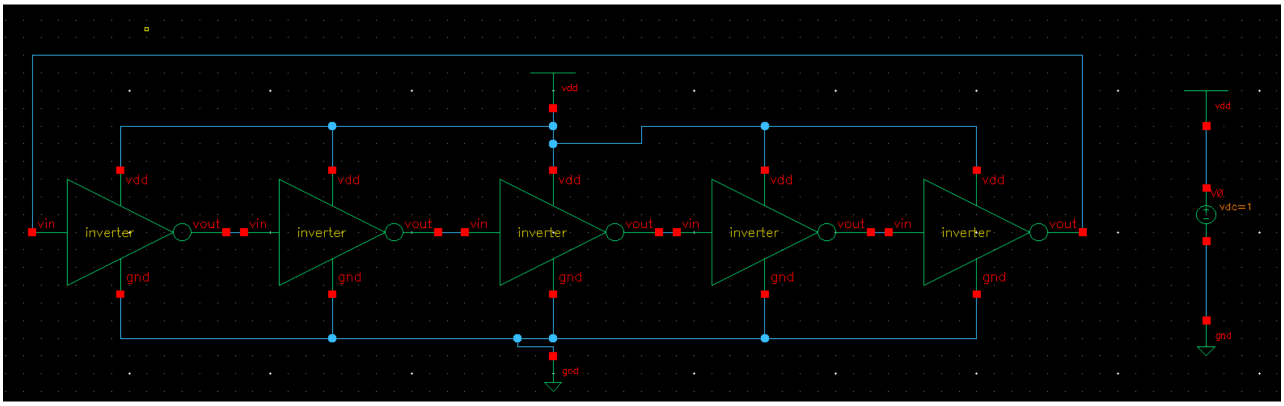

Figure 1.4: Schematic of the Ring Oscillator

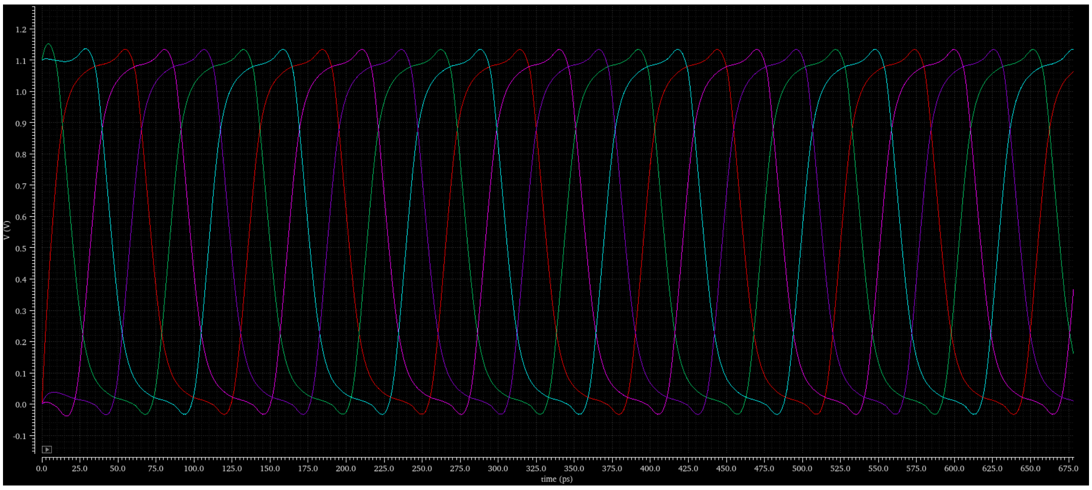

Figure 1.5: Transient Response of the Ring Oscillator

To calculate oscillation frequency, which often depends on supply voltage, transistor sizing, and also load capacitance. the formula is used:

$$f_{\text{osc}} = \frac{1}{2Nt_{pd}}$$

2. Project Description

2.1 Exploration of Complex Polygonal Ring Oscillator Geometry for TRNGs

Advanced ring oscillator topologies that go beyond straightforward linear chains are examined in this section. Designers can create multiple, interacting feedback loops by incorporating the oscillator into intricate geometric layouts such as pentagonal, hexagonal, and tetrahedral structures. In order to create true randomness, this complexity is used to magnify intrinsic noise phenomena like jitter and metastability. The principles and features of these geometries as described in the cited paper are described in detail in the ensuing subsections.

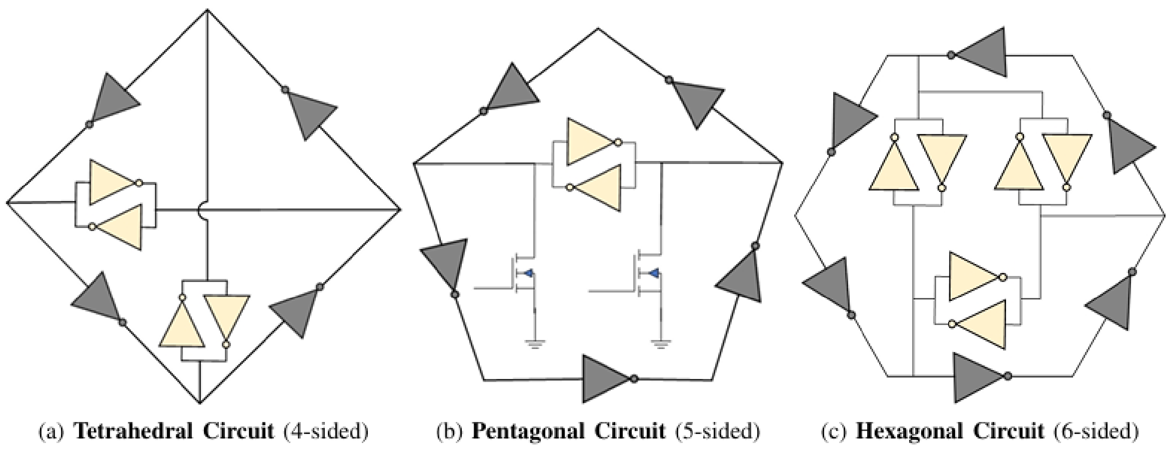

Figure 2.1: Structural Comparison of Ring Oscillator Circuits

2.1.1 Tetrahedral Ring Oscillator

Basic Principle

The Tetrahedral Ring Oscillator isn't just a basic chain of inverters like a standard ring oscillator. Instead, it uses multiple interconnected feedback paths that form a tetrahedral structure. Because of this interconnection, the circuit ends up having four inherently unstable loops and three stable ones. The constant competition between these paths introduces unpredictability into the signal. This natural instability is useful because it increases randomness and provides a stronger entropy source, making the circuit suitable for generating true random numbers rather than predictable or pseudo-random outputs.

Core Architecture and Operation

- **Inverter Count:** Utilizes 8 inverters to form its 4-sided (tetrahedral) structure.
- **Entropy Source:** Relies on the accumulation of timing jitter across its multiple inverter stages and the complex interaction between its loops.
- **TRNG Mechanism:** The high number of interconnected loops within the ring oscillator within the four sided topology contribute to a high-gain, noise-sensitive environment. When sampled, the jitter from these oscillations provides a stream of random bits.

Characteristics

- **Complexity:** High structural complexity with 7 total loops (4 unstable, 3 stable).
- **Performance:** Serves as a benchmark for TRNG designs but is outperformed in area and power efficiency by the pentagonal design.

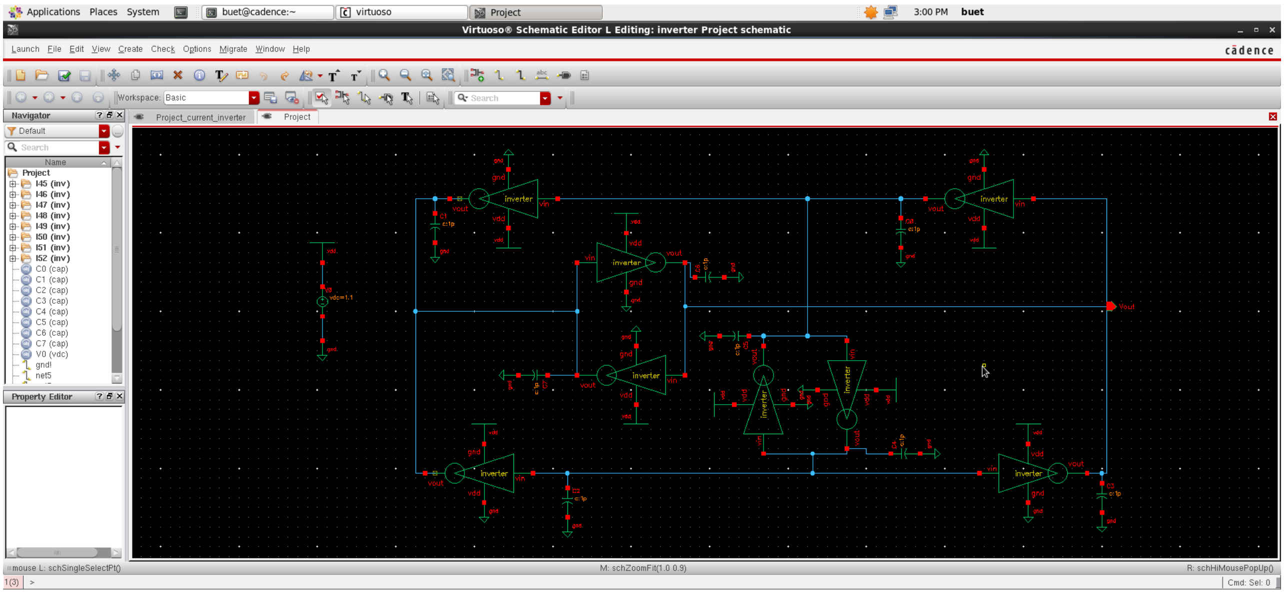

Figure 2.2: Schematic Diagram of Hexagonal Ring Oscillator

2.1.2 Pentagonal Ring Oscillator

Basic Principle

The Pentagonal Ring Oscillator (PRO) is a five-node variation of the traditional ring oscillator that aims to strike a balance between simplicity and performance for TRNG designs. By arranging the feedback network in a pentagon rather than a straight loop, the circuit naturally enters regions of jitter and occasional metastability. These effects are not just noise—they are what make the generated output unpredictable. Because of this design choice, the PRO can produce strong random bitstreams while still keeping the hardware requirements relatively low.

Core Architecture and Operation

- This design prioritizes efficiency by using only seven inverters, which is fewer compared to tetrahedral or hexagonal architectures. Among these seven, two inverters are paired with NMOS transistors to form a metastability core. When the clock signal momentarily disables the NMOS devices, the circuit enters an unstable state where minimal intrinsic noise forces two internal nodes (V_2 and V_5) to settle into opposite logic levels.

- After this metastable phase resolves, the remaining chain of five inverters begins free oscillation. Due to thermal noise and device-level variations, the oscillation exhibits noticeable timing jitter, which becomes a natural source of randomness.

- To extract usable random data, a D-Flip Flop samples the oscillator output. Each sampling event captures a noise-influenced moment in the signal, producing a digital bitstream that behaves unpredictably and is suitable for true random number generation.

Characteristics

- **Efficiency:** Superior performance in area, power, and energy per bit compared to tetrahedral and hexagonal counterparts.

- **Entropy Combination:** Uniquely combines the powerful entropy of a metastable start-up state with the continuous entropy of oscillation jitter.

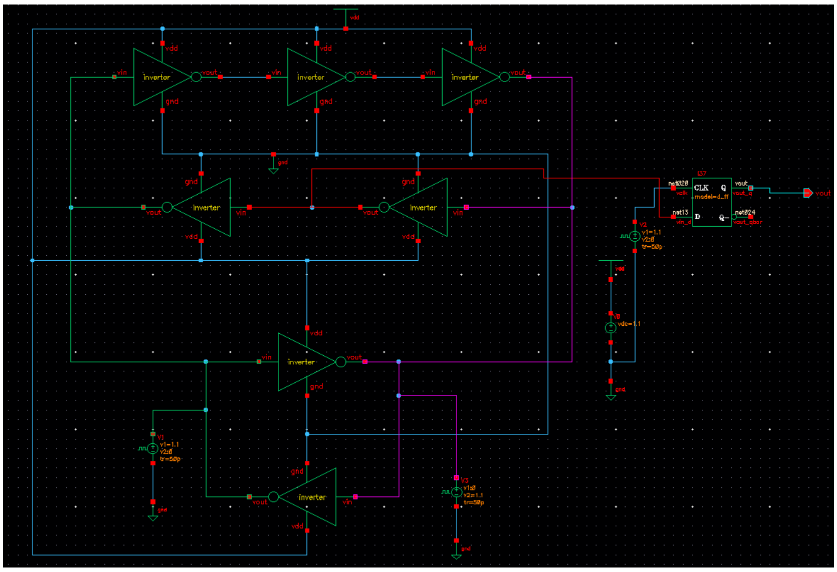

Figure 2.3: Schematic of the proposed Pentagonal Ring Oscillator

2.1.3 Hexagonal Ring Oscillator

Basic Principle

The Hexagonal Ring Oscillator is a larger version with a 6-sided topology that presents a high number of interacting loops. Although quite effective for generating entropy, its larger size compared to the pentagonal design makes it less efficient in key performance metrics.

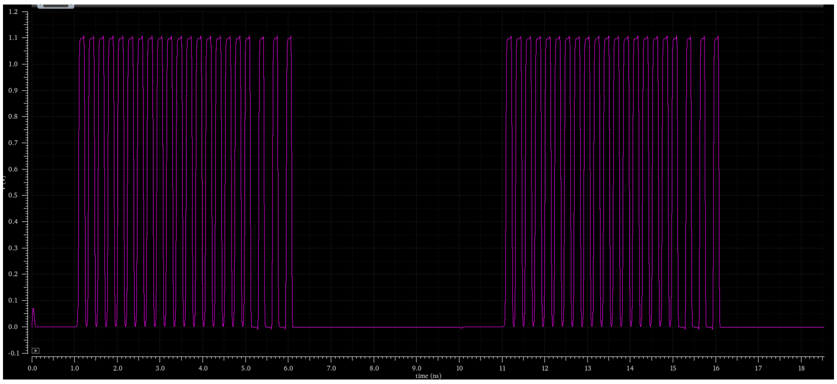

Figure 2.4: Transient Response of Pentagonal Ring Oscillator

Core Architecture and Operation

- **Inverter Count:** with largest geometry it has 12 inverters..
- **Entropy Source:** Its working is very similar to a tetrahedral oscillator but deals with broader scale. Its oscillation is fueled by the odd-inversion rule in its primary loop, and jitter is collected over many stages.
- **TRNG Mechanism:** The extensive network of 15 total loops (8 unstable, 7 stable) contributes to a complex jitter profile that can be sampled for randomness.

Characteristics

- **Scale:** Has the highest number of components and loops, leading to the largest layout area.
- **Performance:** Generally results in lower oscillation frequency and higher power consumption compared to more compact designs like the PRO, due to the increased capacitive load and number of stages.

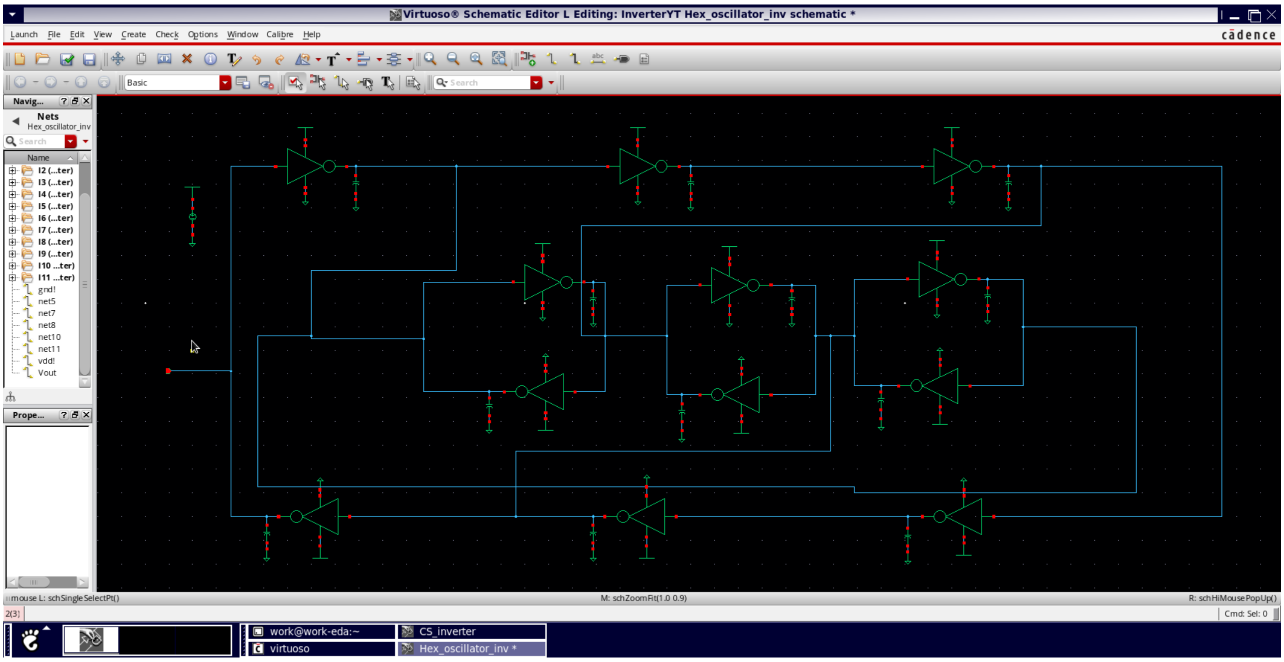

Figure 2.5: Schematic Diagram of Hexagonal Ring Oscillator

2.1.4 Comparative Summary

Table 2.1: Comparison of Ring Oscillator Topologies for TRNGs

Oscillator Type	**Inverters**	**Key Entropy Source**	**Relative Efficiency**
Tetrahedral	8	Jitter in multiple loops	Medium
Pentagonal (PRO)	**7**	**Metastability + Jitter**	**High**
Hexagonal	12	Jitter in extensive loops	Low

2.2 Ring Oscillator Comparison

Parameter	Tetrahedral (4)	Pentagonal (5)	Hexagonal (6)
Stages	4 (Even)	5 (Odd)	6 (Even)
Principle	Cross-coupling	Natural odd-stage	Phase compensation
Phase Shift	Complex distributed	$5 \times 216°$	$1080° + \phi_{extra}$
Frequency	Medium ($1/8t_d$)	Low ($1/10t_d$)	Lowest ($1/12t_d$)
Output Phases	4 (90°)	5 (72°)	6 (60°)
Complexity	High	Low	Medium
Stability	Medium	High	High
Phase Noise	Medium	Low	Lowest
Power	High	Medium	High
Applications	Research, Edu	Clock generation	Multi-phase systems
Reliability	Low	High	Medium
Area	Large	Compact	Medium

Table 2.2: Comparison of ring oscillator configurations

2.3 Design and Implementation of the Current-Starved CMOS Ring Oscillator

The Current-Starved Ring Oscillator (CSRO) works by limiting the charging and discharging currents of every inverter stage using an extra series-connected NMOS and PMOS transistors. These current-limiting devices act as VCCs, allowing the propagation delay of each and every inverter to be adjusted using the control voltage V_{ctrl}. This tunability makes the CSRO primarily different from a basic CMOS ring oscillator, where we fix the delay by device sizing and amount of current supply. By increasing V_{ctrl}, more current is now allowed to flow, which finally reduces delay and increases oscillation frequency; reducing V_{ctrl} has the vive versa effect. This architecture enables more stable,more controlled, and efficient powered oscillation suitable for VCOs, PLLs, and TRNG applications.

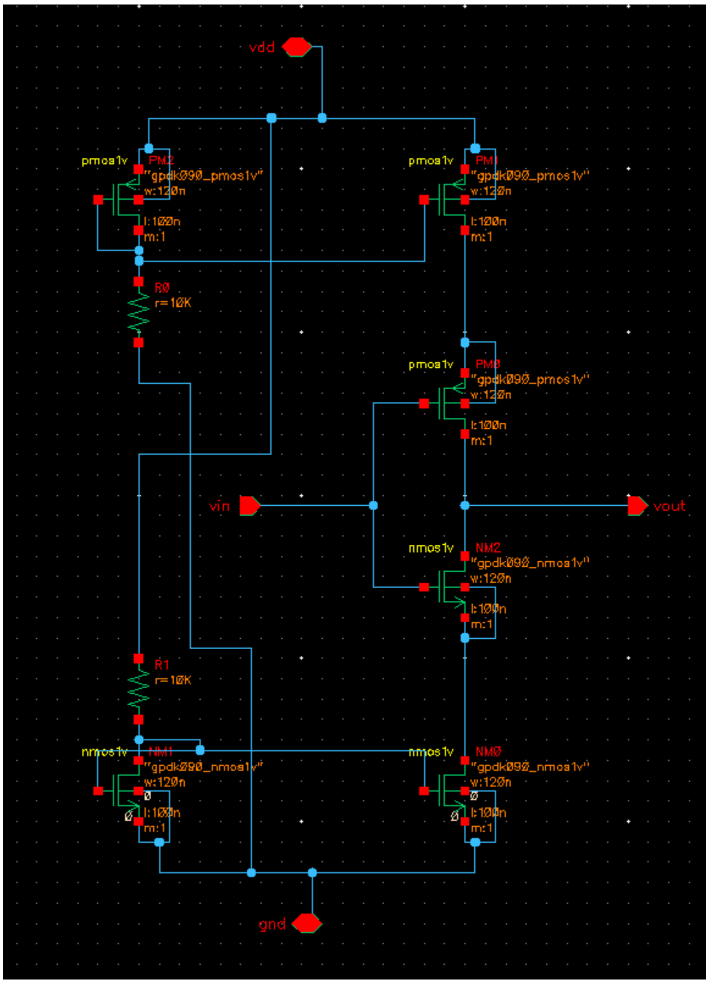

Figure 2.6: Schematic of the Current-Starved CMOS Ring Oscillator

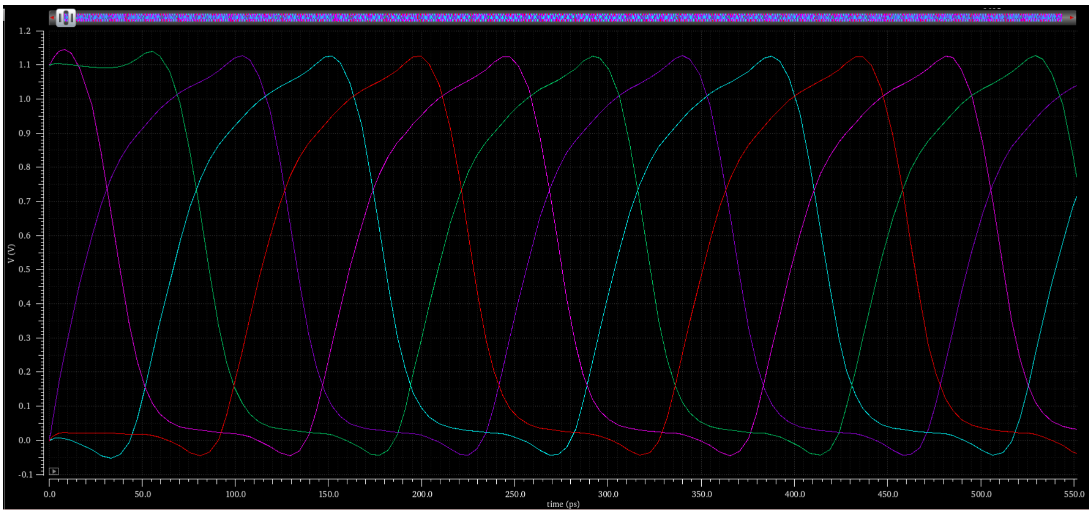

Figure 2.7: Output Plots of the Current-Starved CMOS Ring Oscillato

Analysis: The oscillation frequency of the CSRO is calculated by both the propagation delay of the inverter and the current-limiting action of all the control transistors. Due to restricted current availability, the CSRO generally produces a lower oscillation frequency as compared to a normal CMOS ring oscillator that is operating at the same supply voltage. A crucial advantage of this structure is that the output waveform exhibits a lower Total Harmonic

Distortion (THD), as the controlled current results in smoother transitions and reduced high-frequency noise components. The CSRO gives tunable frequency control,and pure signals, best for low noise and low power applications.

2.4 Design and Implementation of the Dynamic CMOS Inverter Ring Oscillator

Dynamic CMOS Inverter:

The division of operation into two separate, clock-controlled phases is the fundamental idea behind dynamic CMOS logic. It is based on a typical static CMOS inverter, however it also includes a**clock (CLK)** signal to control the timing and flow of charge. In the *precharge phase* (CLK = 0), The pull-up network uses a clock-controlled PMOS transistor to charge the output node to a predefined logic level, high. In the ensuing*evaluation phase* when (CLK = 1),Depending on the input logic, the NMOS pull-down network conditionally discharges the precharged output. This makes the system quicker but more susceptible to leakage, noise, and charge-sharing effects because the output is only valid while the stored charge is still on the dynamic node. Dynamic CMOS logic reduces transistor count and improves speed, but it requires precise clocking and careful design to maintain signal integrity.

To achieve a higher oscillation frequency and improve randomness, the ring oscillator was redesigned using a dynamic logic–style CMOS inverter. In this approach, each stage operates with a clocked precharge phase (CLK low) followed by an evaluation phase (CLK high), enabling faster charging and discharging within the same cycle.

Analysis: Dynamic logic reduces the number of transistors in the critical path per stage, minimizing propagation delay. The evaluate phase of one stage automatically triggers the precharge of the next stage, creating a fast, self-timed ring. This structure increases oscillation frequency and increases pseudo-randomness of the o/p compared to all static CMOS and current-starved designs.

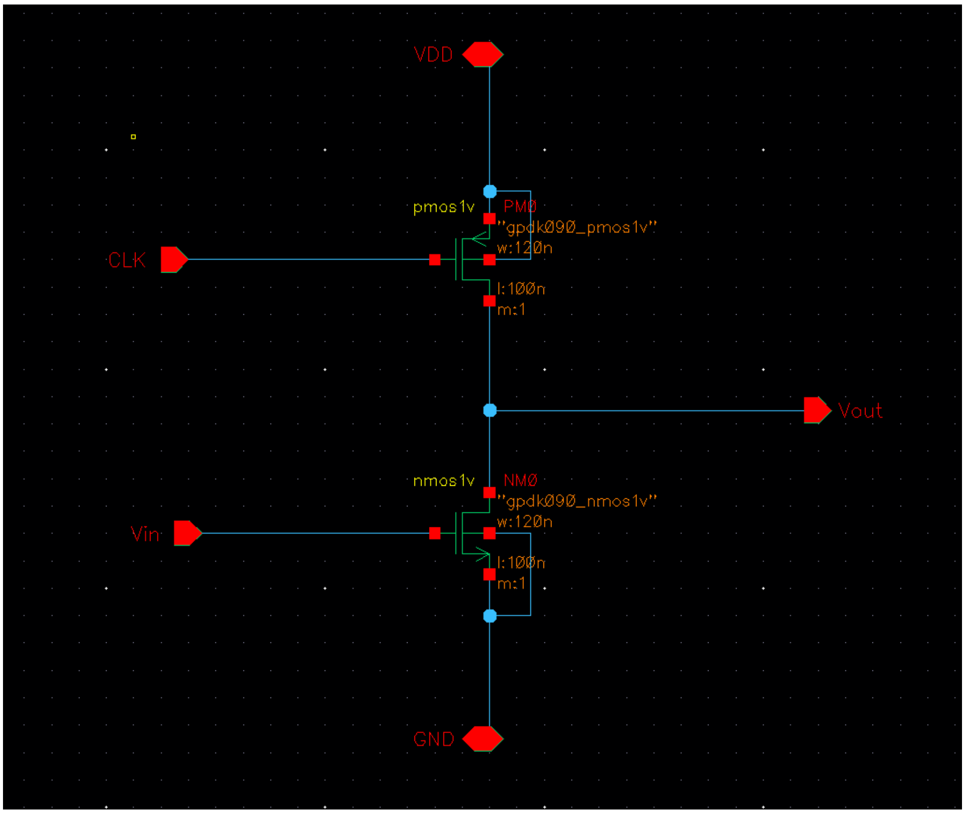

Figure 2.8: Schematic of the Dynamic CMOS Inverter

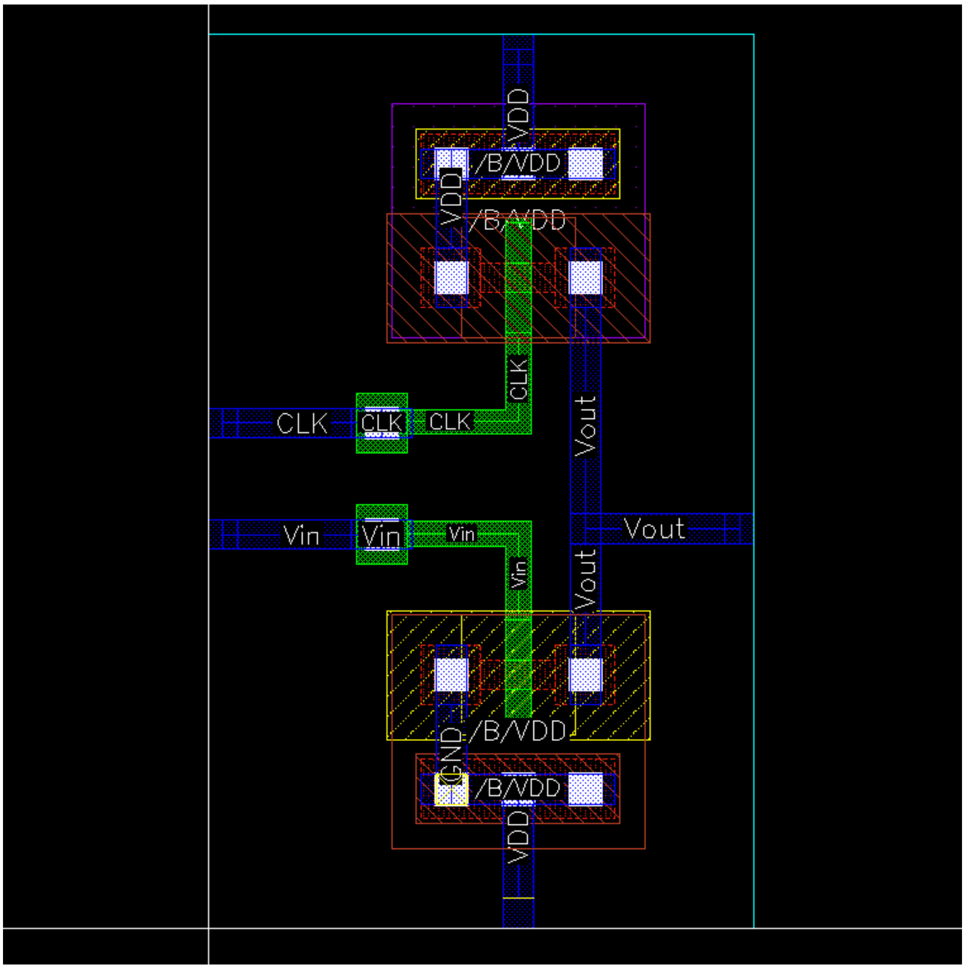

Figure 2.9: Layout of the Dynamic CMOS Inverter

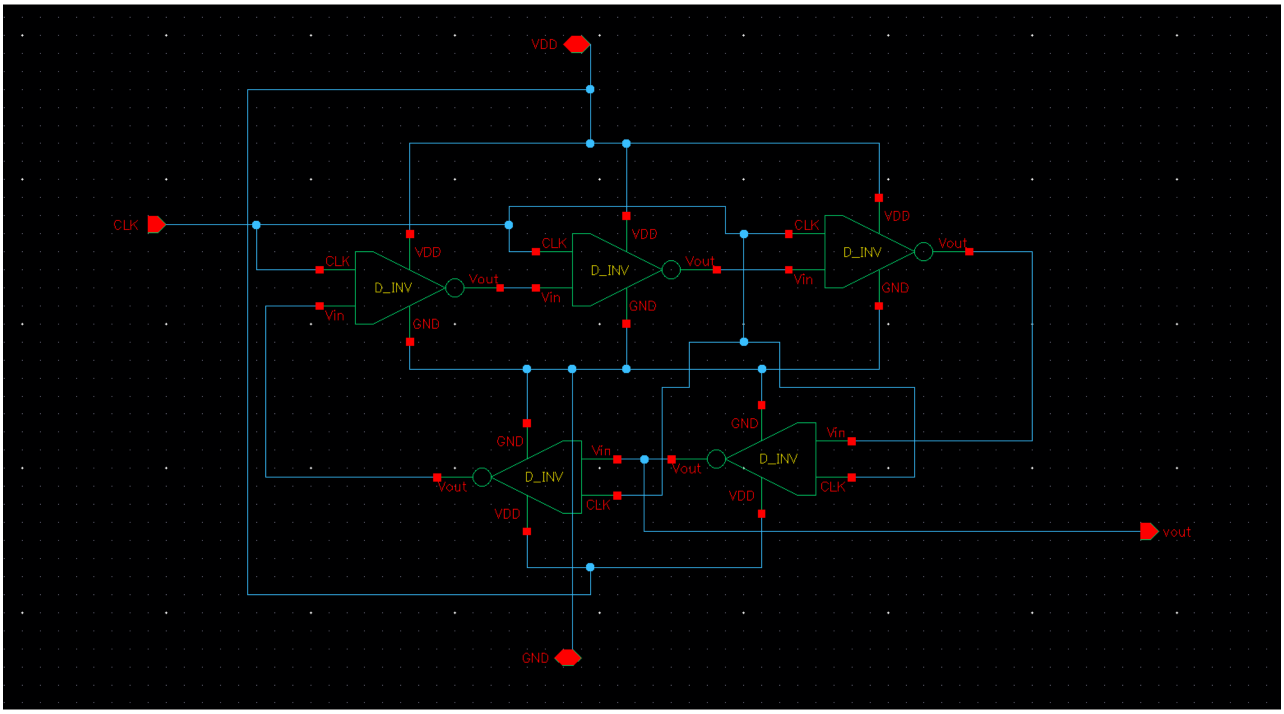

Figure 2.10: Schematic of the Dynamic CMOS Inverter Ring Oscillator

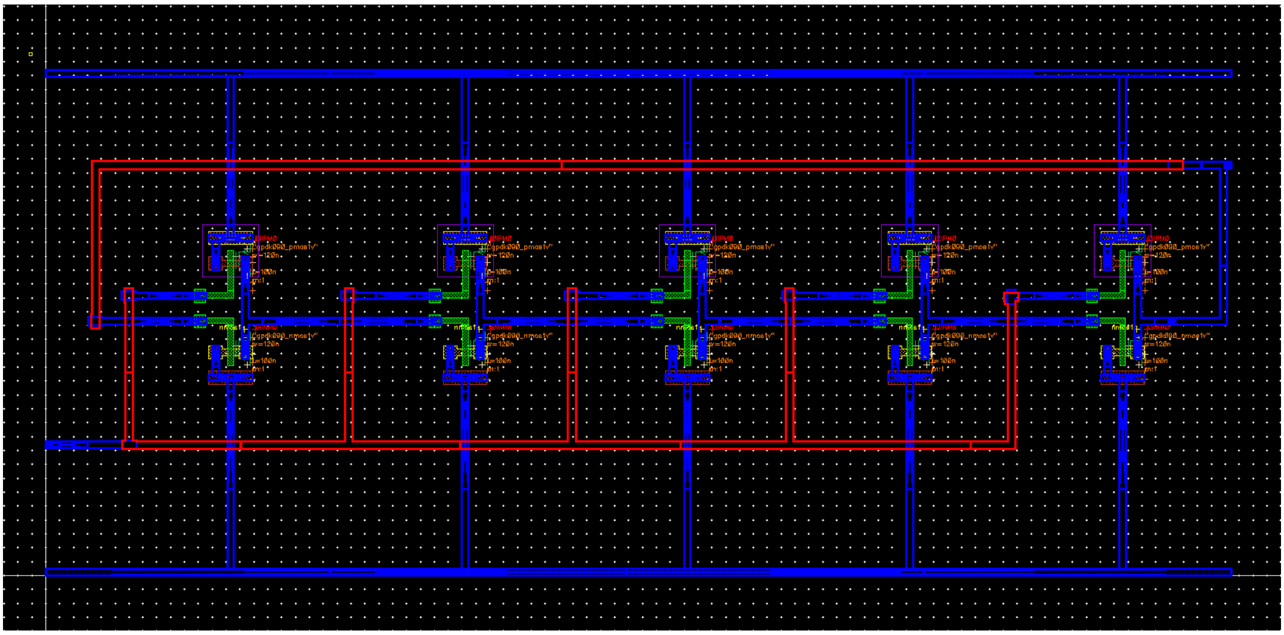

Figure 2.11: RC Extraction of Dynamic CMOS Inverter Ring Oscillator

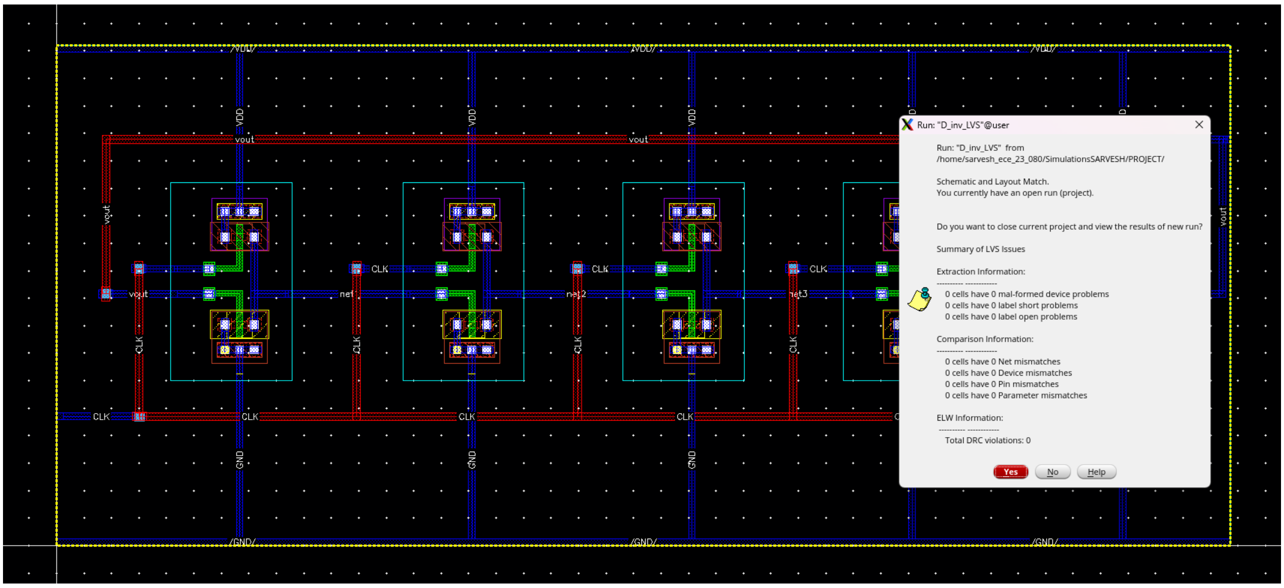

Figure 2.12: LVS of the Dynamic CMOS Inverter Ring Oscillator

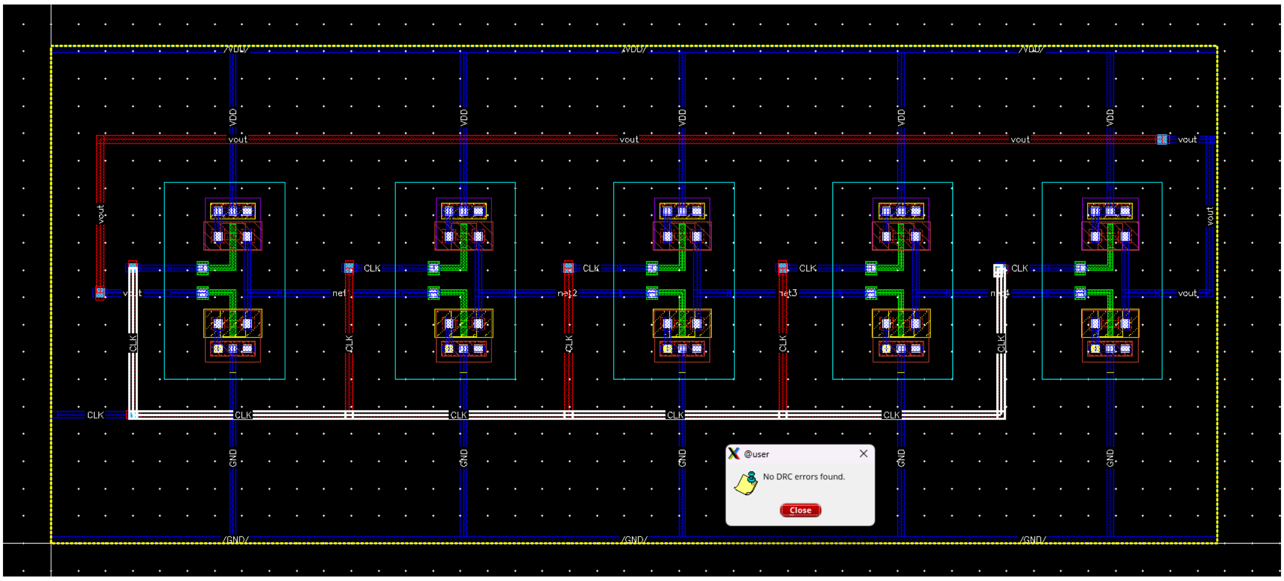

Figure 2.13: DRC of the Dynamic CMOS Inverter Ring Oscillator

The complete working of the Dynamic CMOS Inverter Ring Oscillator starts with the schematic design of the dynamic inverter-based ring structure. After the verification of the schematic , the layout is made with proper transistor placement and interconnections. The RC extraction step catches the real parasitic resistances and capacitances, enabling right post-layout timing analysis. LVS is performed to make sure that the layout matches the schematic perfectely, confirming right functionality. Finally, DRC proves that the layout follows all fabrication rules without any violations. These steps together establish a right and reliable dynamic ring oscillator design.

3. Results

3.1 The analysis of Tetrahedral, Pentagonal, and Hexagonal Ring Oscillator

The starting analysis was performed for tetrahedral, pentagonal, and hexagonal ring oscillators. The changes of frequency, THD, average power, and jitter were observed with respect to the supply voltage V_{DD}. These results assisted in understanding the expandability of oscillator performance with an increased number of stages.

3.2 Current-Starved Inverter Based Ring Oscillator

We have designed and simulated a current-starved inverter–based ring oscillator successfully. In the Current starved by limiting the amount current through the transistor we are controlling the frequency of the oscillation and hence controlling the amount of jitter. The same sizing as the CMOS inverter, this topology provided the following results:

- Lower oscillation frequency
- Reduced noise
- Higher delays due to restrictions on current during charging and discharging

This verifies the power–performance trade-off in current-starved designs.

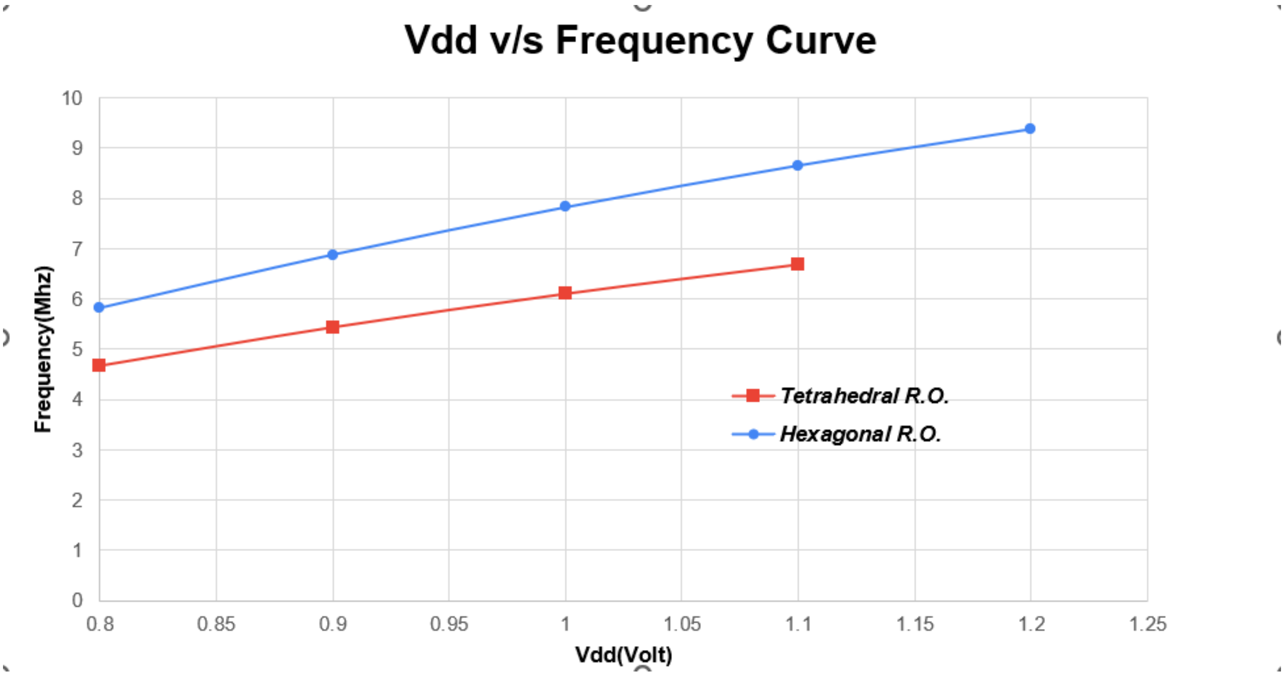

Figure 3.1: V_{DD} vs Frequency for Tetrahedral and Hexagonal Ring Oscillators

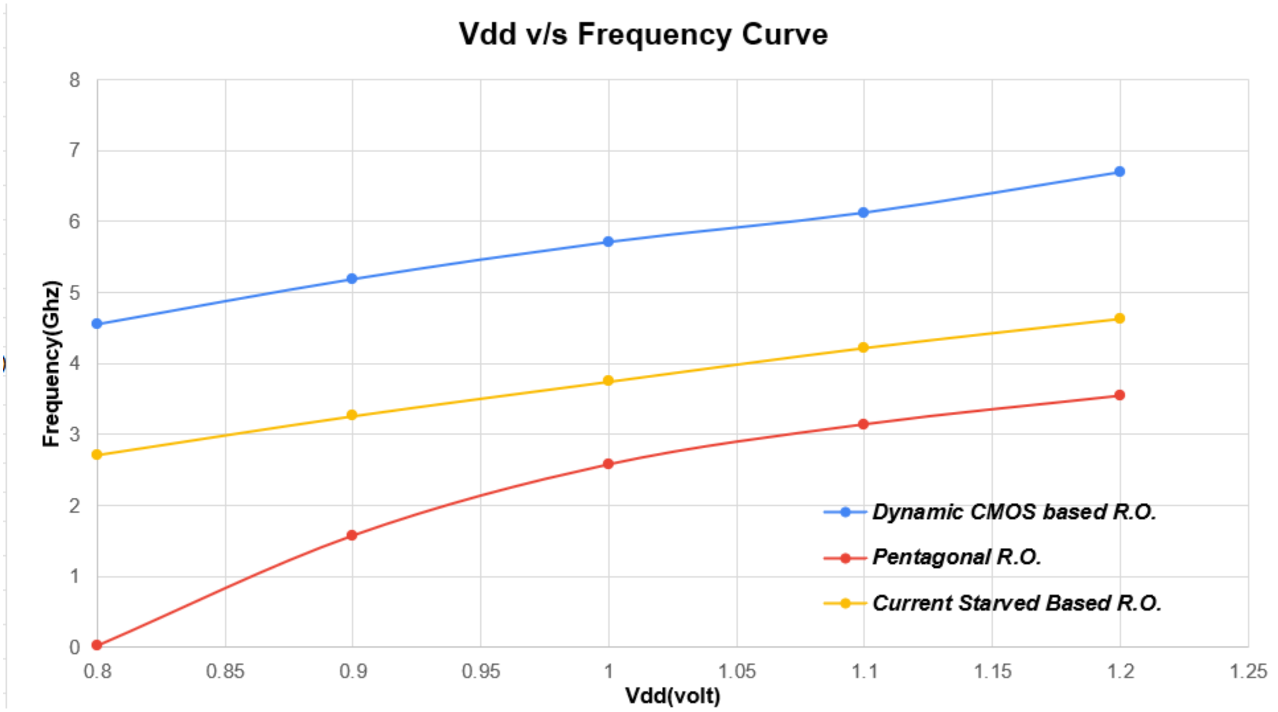

Figure 3.2: V_{DD} vs Frequency for Pentagonal, Dynamic CMOS, and Current-Starved Ring Oscillators

Frequency Comparison: The oscillation frequency is highest in the Dynamic CMOS ring oscillator due to its precharge–evaluate mechanism, followed by the Tetrahedral and Hexagonal oscillators. The Pentagonal (5-stage) oscillator exhibits the lowest frequency due to its higher stage delay. This trend highlights the combined effects of inverter type, stage count, and propagation delay on the overall oscillation frequency.

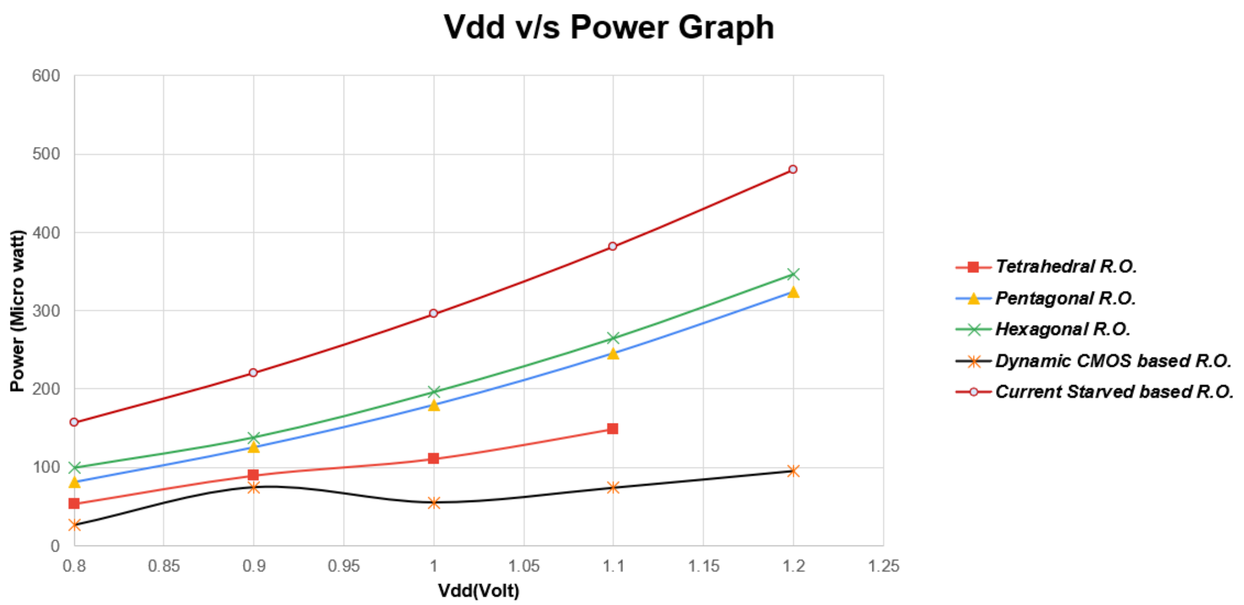

Figure 3.3: V_{DD} vs Power for Tetrahedral, Hexagonal, Pentagonal, Dynamic CMOS, and Current-Starved Ring Oscillators

Power Comparison: The average power consumption of the ring oscillators in ascending order is Dynamic CMOS, Tetrahedral (4-stage), Pentagonal (5-stage), Hexagonal (6-stage), and Current-Starved. This order reflects the combined effects of stage count, inverter type, and switching activity on power dissipation.

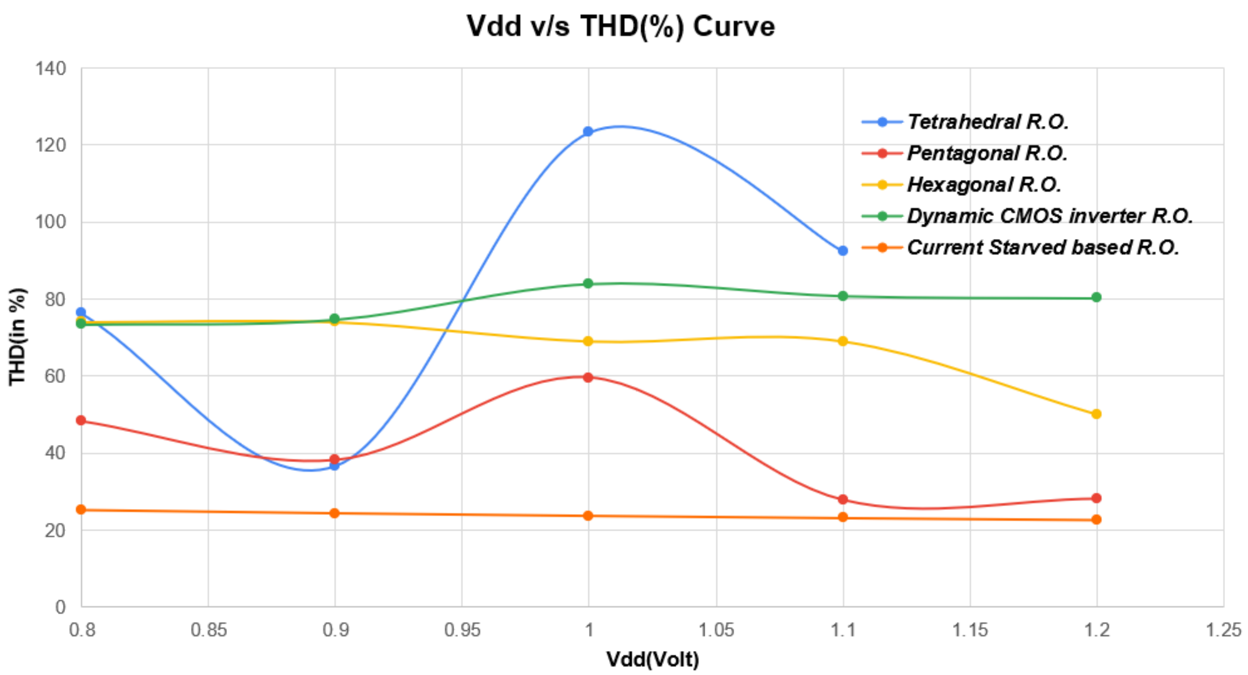

Figure 3.4: V_{DD} vs THD (in Percentage) for Tetrahedral, Hexagonal, Pentagonal, Dynamic CMOS, and Current-Starved Ring Oscillators

THD Comparison: The total harmonic distortion (THD) is highest in the Tetrahedral (4-

stage) oscillator, but it is unstable. Following this, the THD decreases in the Dynamic CMOS oscillator, Hexagonal (6-stage), Pentagonal (5-stage), and Current-Starved ring oscillators. This trend reflects the impact of stage configuration and inverter type on signal distortion and overall stability.

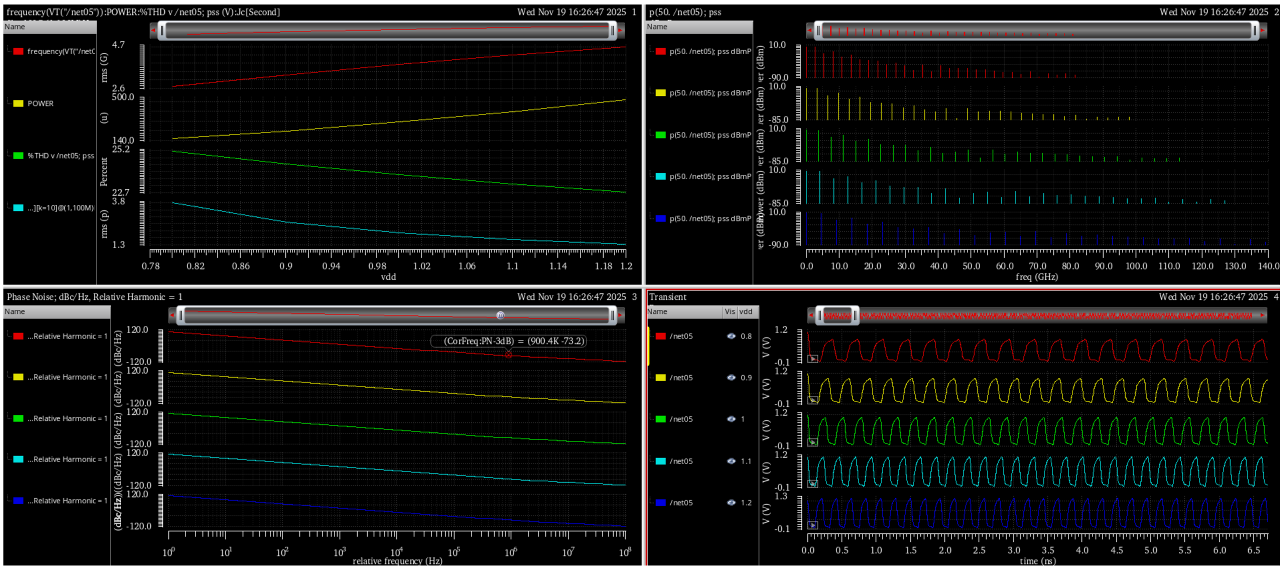

Figure 3.5: Performance Metrics of the Current-Starved Inverter Ring Oscillator

3.3 Dynamic CMOS Inverter Based Ring Oscillator

The dynamic CMOS inverter–based ring oscillator was implemented next. This architecture demonstrates:

- Higher oscillation frequency
- Higher noise due to dynamic node switching
- Lower average power consumption due to charge-storage behavior

Dynamic logic thus enables high-speed operation at reduced power but with increased sensitivity to noise.

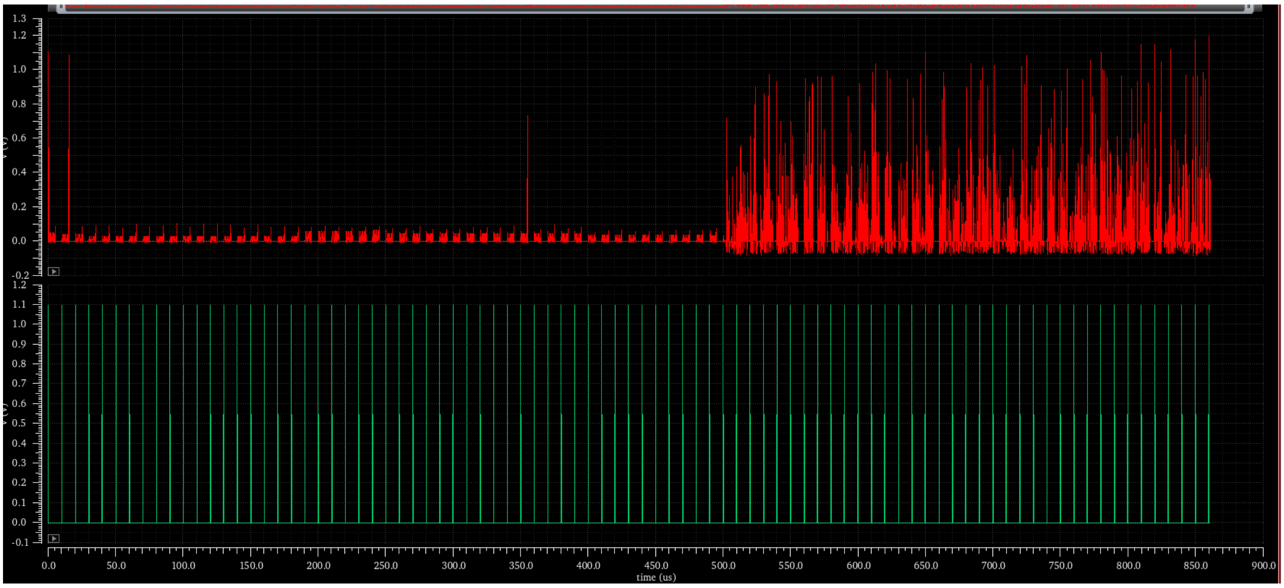

Figure 3.6: Random bit generated by Dynamic CMOS Inverter based Ring Oscillator

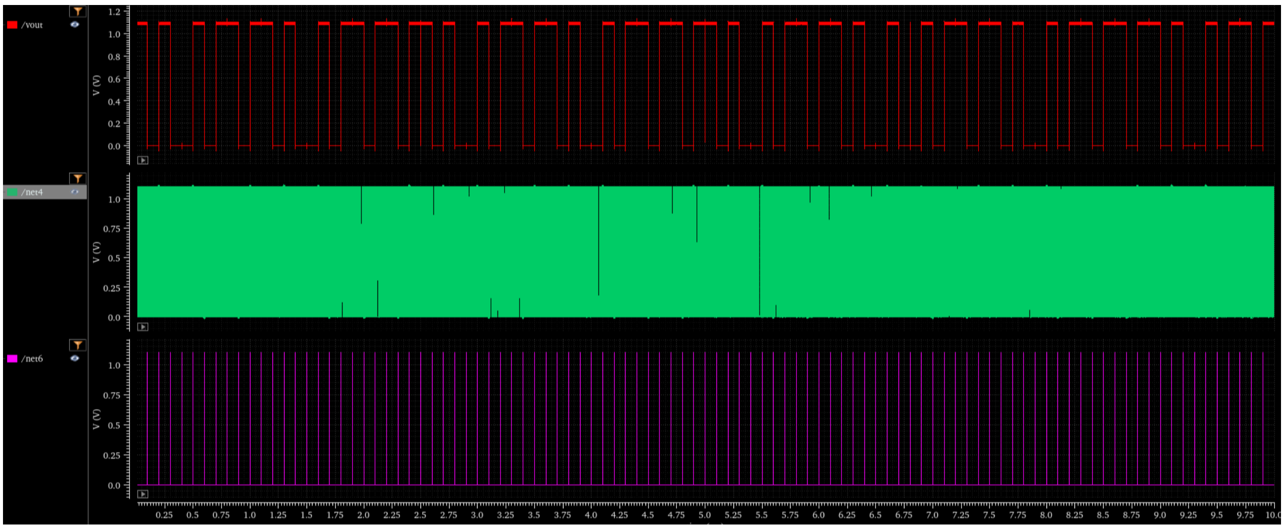

Figure 3.7: Random bit generated by Dynamic CMOS Inverter based Ring Oscillator

4. Conclusion

Key conclusions:

- **Geometric Topology Analysis:** Different performance characteristics were found when complicated feedback architectures, namely Tetrahedral, Pentagonal, and Hexagonal oscillators, were compared. Although the hexagonal architecture provides a large number of loop interactions, capacitive loading causes it to have a larger area and a lower frequency.The **Pentagonal Ring Oscillator (PRO)** became the ideal geometric balance, offering better entropy through jitter and metastability while retaining higher area efficiency than its hexagonal and tetrahedral cousins..
- **Current-Starved Architecture:** The -operated and extremely low-power security systems are best suited for a Current-Starved Ring Oscillator (CSRO). It provides robust noise reduction and tunable frequency control by limiting the charging and discharging currents. However, compared to standard inverter-based oscillators, this power efficiency comes at the expense of a lower oscillation frequency.
- **Dynamic CMOS Performance:** The the highest oscillation frequency and surprisingly low power consumption, the Dynamic CMOS Inverter-based RO demonstrated the best overall performance. Its sensitivity to layout parasitics and charge-sharing, which is typically a disadvantage, actually reduced jitter and noise, making it ideal for high-entropy random number generation.
- **Physical Implementation & Robustness: full-custom layouts**, verified through **Design Rule Checking (DRC)** and designs were verified to be ready for manufacture through Layout Versus Schematic (LVS) inspections. The static CMOS inverter rings demonstrated the strongest stability against process fluctuations, offering a robust baseline.

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